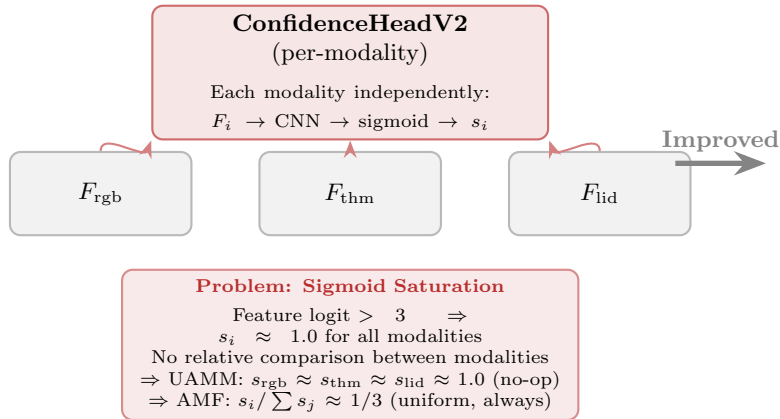
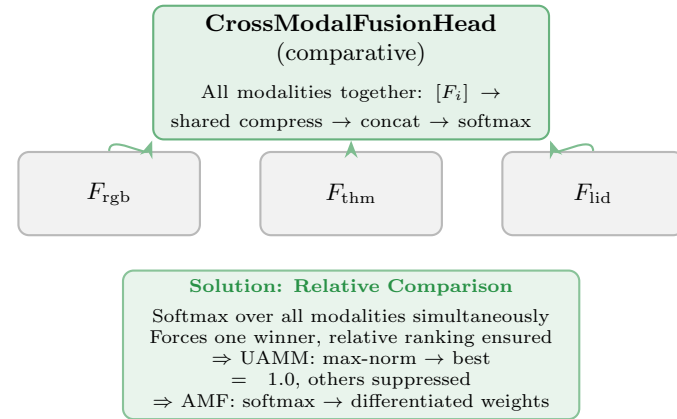


P8 → P9: Key Improvements

P8 (Problem)



P9 (Solution)



Effect on UAMM Scores

P8 UAMM (sigmoid):

Night scene: $s_{\text{rgb}} = 0.98$,
 $s_{\text{thm}} = 0.99$, $s_{\text{lid}} = 0.97$
 All near 1.0 $\rightarrow$ no modality suppressed
 $\rightarrow$ degraded RGB contaminates output

VS.

P9 UAMM (max-norm):

Night scene: $\hat{w}_{\text{rgb}} = 0.40$,
 $\hat{w}_{\text{thm}} = 1.00$, $\hat{w}_{\text{lid}} = 0.60$
 Thermal dominates $\rightarrow$ RGB suppressed
 $\rightarrow$ clean thermal-guided prediction

P9 Components: (1) Soft-MoE LoRA $\times$ 3 modalities (2) CrossModal-FusionHead (shared compress + compare MLP) (3) UAMM (max-norm) (4) AMF (softmax fusion)

Training Loss: $\mathcal{L} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{proto}}$ No oracle supervision, no MI loss — purely data-driven.